Appendix A: Preprocessing Steps

CTG Download. We downloaded the CTG results for completed studies with posted results as a JSON export (all available fields) from `https://clinicaltrials.gov/search?aggFilters=results:with,status:com` on August 7th, 2025. During data cleaning, we excluded four records with sample-size errors identified while reviewing candidate type I errors at $\alpha = .01$.

Record Construction. We transformed each ClinicalTrials.gov JSON record into flat, analysis-level rows prior to p-value cleaning and method harmonization. For each study (`nctId`), we iterated over outcome measures with at least one analysis and, for every outcome–analysis pair, emitted one row per denominator count with units "Participants" and with `groupId` in the analysis's declared `groupIds`. We carried forward study metadata (e.g., `briefTitle`, `studyType`), design-information fields (`interventionModel`, `observationalModel`, `timePerspective`, `allocation`), one-hot encoded phases (`EARLY_PHASE1`, `PHASE1`, `PHASE2`, `PHASE3`, `PHASE4`), and MeSH terms/IDs for conditions and interventions from the `derivedSection`. We also flagged whether the study included any references of type `RESULTS` or `DERIVED`.

Analysis Identifier. Each row was keyed by a stable, hash-based identifier,

$$\texttt{analysisId} = \mathrm{MD5}\big(\texttt{nctId} + \mathrm{JSON}(\texttt{outcome};\ \text{sorted keys}) + \mathrm{JSON}(\texttt{analysis};\ \text{sorted keys})\big),$$

which yields reproducible IDs for unique outcome–analysis pairs within a study (independent of JSON key order). This `analysisId` serves as the unit of aggregation in downstream preprocessing (e.g., computing sample sizes and $eJAB$).

Schema. Fields include: `analysisId`, `nctId`, `briefTitle`, `outcomeType`, `outcomeTitle`, `groupDescription`, `units`, `groupId`, `value`, `pValue`, `statisticalMethod`; MeSH fields (`condition_mesh`, `condition_ids`, `intervention_mesh`, `intervention_ids`); `hasResultsOrDerived`; study-design fields; and phase indicators. The field `value` records the number of participants for the given outcome group/arm.

Standardization of Reported p-Values. Reported p-values often included textual qualifiers (e.g., "<0.05"). We standardized them as follows:

1. Flag the presence of a "<" qualifier: `has_less_than` ← `grepl("<", pValue)`.
2. Extract the numeric component via regex:

$$\texttt{pValue} \leftarrow \texttt{str_extract(pValue, "[0-9]*.?[0-9]+")}.$$

3. Drop rows with missing or blank numeric components.
4. Coerce `pValue` to numeric and retain only valid probabilities: $0 <$ `pValue` < 1.
5. Exclude rows with zero sample size, which are incompatible with downstream model mappings.

Harmonization of Statistical Method Names. Free-text `statisticalMethod` labels were mapped to canonical test families using regular-expression rules (`str_detect`). Observations with no match were excluded. The exact mapping rules are:

Listing 1: Regex-based mapping of free-text methods to canonical families

```
statisticalMethod <- case_when(
  # --- T-TESTS ---
  str_detect(originalMethod, regex("\\bt[- ]?test", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("one[- ]?sample", ignore_case = TRUE)) ~ "1-ttest",

  str_detect(originalMethod, regex("\\bt[- ]?test", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("paired", ignore_case = TRUE)) ~ "1-ttest",

  str_detect(originalMethod, regex("\\bt[- ]?test", ignore_case = TRUE)) ~ "2-ttest",

  # --- CONDITIONAL LOGISTIC REGRESSION ---
  str_detect(originalMethod, regex("logistic", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("regression", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("conditional", ignore_case = TRUE)) ~ "clogit",

  # --- LOGISTIC REGRESSION ---
```

```
  str_detect(originalMethod, regex("logistic", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("regression", ignore_case = TRUE)) ~ "glm",

  # --- COX ---
  str_detect(originalMethod, regex("cox", ignore_case = TRUE)) &
    !str_detect(originalMethod, regex("wilcoxon|mantel|signed|rank", ignore_case = TRUE)) ~ "cox",

  # --- LOGRANK / MANTEL-COX ---
  str_detect(originalMethod, regex("log ?rank|mantel[- ]?cox", ignore_case = TRUE)) &
    !str_detect(originalMethod, regex("wilcoxon|signed", ignore_case = TRUE)) ~ "logrank",

  # --- MANN-WHITNEY ---
  str_detect(originalMethod, regex("mann[- ]?whitney", ignore_case = TRUE)) |
    (str_detect(originalMethod, regex("wilcoxon", ignore_case = TRUE)) &
       str_detect(originalMethod, regex("rank", ignore_case = TRUE)) &
       !str_detect(originalMethod, regex("signed", ignore_case = TRUE))) ~ "Utest",

  # --- WILCOXON SIGNED-RANK ---
  str_detect(originalMethod, regex("wilcoxon", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("signed", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("rank", ignore_case = TRUE)) ~ "wilcox",

  # --- KRUSKAL-WALLIS ---
  str_detect(originalMethod, regex("kruskal|wallis", ignore_case = TRUE)) ~ "Htest",

  # --- CHI-SQUARE ---
  str_detect(originalMethod, regex("chi[- ]?square|χ2|chi2|chisq", ignore_case = TRUE)) ~ "chisq",

  # --- REPEATED MEASURES ---
  str_detect(originalMethod, regex("repeated[- ]?measures?", ignore_case = TRUE)) ~ "rANOVA",

  # --- LINEAR REGRESSION ---
  str_detect(originalMethod, regex("linear", ignore_case = TRUE)) &
    str_detect(originalMethod, regex("regression", ignore_case = TRUE)) ~ "lm",

  # --- ANOVA ---
  str_detect(originalMethod, regex("anova|analysis of variance", ignore_case = TRUE)) ~ "ANOVA",

  TRUE ~ NA_character_
)
```

Adjustment for One-Sided p-Values. For explicitly one-sided scalar analyses, we reconstructed the two-sided p-value as

$$p = 2\min\{p_1, 1 - p_1\}.$$

The null-centered one-sided eJAB omits $(n-1)/n$ from the exponent and uses the directional divisor $2(1-p_1)$. NHST instead requires $2p_1 \leq \alpha$. Appendix C.5 gives the method-label rules and full formula.

analysisId Sample Size QC. Certain tests should yield exactly one row per `analysisId` (e.g., one-sample t-test, Wilcoxon signed-rank test). To enforce consistency, we dropped analyses in which such tests appeared with multiple rows under the same `analysisId`.

Aggregation to the Test Level and Parameter Construction. We aggregated records at the `analysisId` (test) level. Fields invariant within a test were taken from the first row—e.g., `nctId`, `pValue`, `statisticalMethod`, `studyType`, phase indicators (`EARLY_PHASE1`, `PHASE1`, `PHASE2`, `PHASE3`, `PHASE4`), `outcomeTitle`, `outcomeType`, `groupDescription`, `conditionIds`, `briefTitle`, and `has_less_than`.
We defined the effective sample size n per test as:

$$n = \begin{cases} \texttt{first(value)}, & \text{one-sample } t\text{-test or Wilcoxon signed-rank;} \\ \max \texttt{value}, & \text{conditional logistic regression;} \\ 0.5 \sum \texttt{value}, & \text{Cox PH or log-rank;} \\ \sum \texttt{value}, & \text{otherwise.} \end{cases}$$

We also set method-specific parameters required by *eJAB*:

$$\texttt{I} = \begin{cases} n_{\text{rows within test}}, & \text{ANOVA, Kruskal–Wallis, repeated measures;} \\ \text{NA}, & \text{otherwise,} \end{cases}$$

$$(\texttt{R}, \texttt{C}) = \begin{cases} (n_{\text{rows within test}},\ 2), & \text{chi-squared test;} \\ (\text{NA},\ \text{NA}), & \text{otherwise.} \end{cases}$$

Mapping to JAB Models and Computation of eJAB. We mapped canonical `statisticalMethod` labels to the `JABO1` model strings required for $eJAB_{01}$. Mapping was performed only for tests included in our analysis (i.e., excluding chi-squared, Cox PH, log-rank, and repeated-measures designs):

Two-sample t-test, paired t-test, one-sample t-test $\rightarrow$ `"t-test"`,
Linear regression $\rightarrow$ `"linear_regression"`,
Logistic regression, conditional logistic regression $\rightarrow$ `"logistic_regression"`,
ANOVA $\rightarrow$ `"anova"`, Kruskal–Wallis $\rightarrow$ `"kruskal_wallis"`,
Mann–Whitney $\rightarrow$ `"mann_whitney"`, Wilcoxon $\rightarrow$ `"wilcoxon"`.

Given $(n, \texttt{pValue}, \texttt{model}, R, C, I)$ for each test, we computed $eJAB_{01}$ via

$$\texttt{JAB} = \texttt{JABO1}(n,\ \texttt{pValue},\ \texttt{model};\ R,\ C,\ I).$$

Exclusions and Final Dataset. We excluded rows with missing `model`, missing `JAB` or `pValue`, and tests with $n = 1$. The manuscript analysis then applied the criteria in Appendix C to obtain the same screened set used throughout: 16,563 eligible primary- and secondary-outcome analyses from 1,790 exact Early Phase I, Phase I, and Phase II trials.

Appendix B: Supplementary Figures

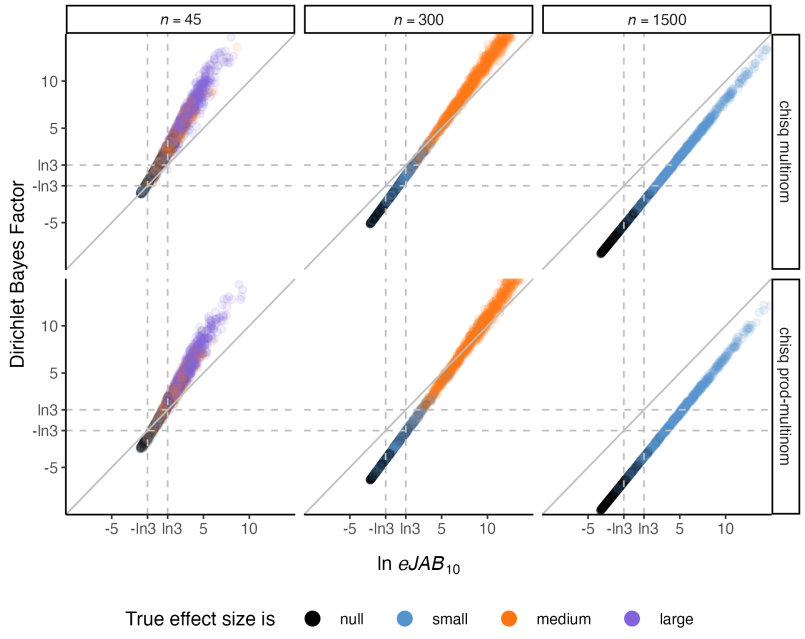


Figure S2: Comparison of the natural logarithmic $eJAB_{10}$ with the `contingencyTableBF` function from the '*BayesFactor*' `R` package at three total counts for chi-squared tests of independence under two sampling designs: multinomial (top) and product-multinomial (bottom). Colors represent the true effect size. The dashed lines indicate $eJAB_{10}$ of 1/3 and 3, representing moderate evidence against and for an effect, respectively.

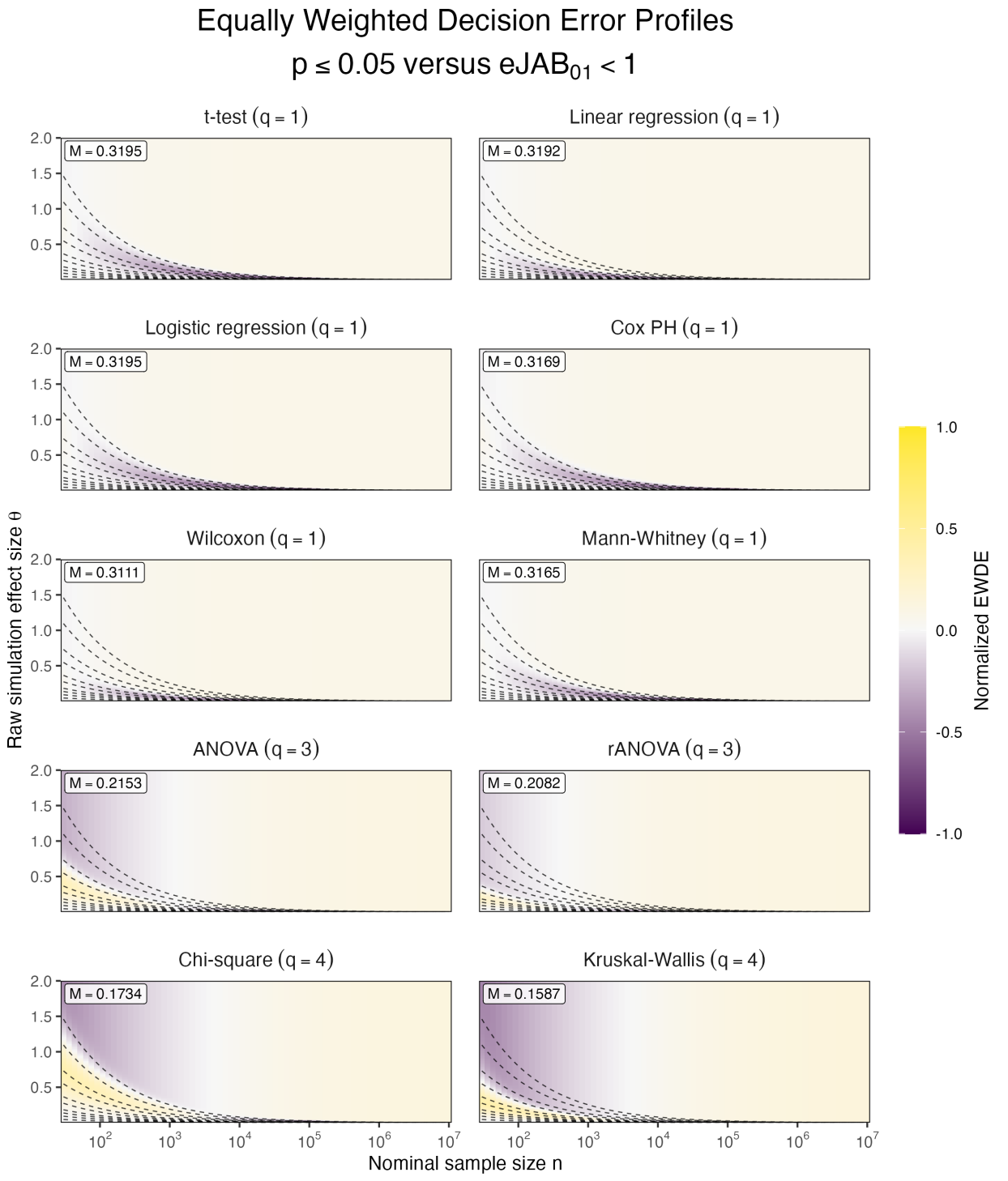


Figure S3: Equally Weighted Decision Error Profiles comparing the NHST significance rule $p \leq .05$ with $eJAB_{01} < 1$ across ten statistical tests. EWDE is the equally weighted average of the type I and type II error probabilities. Colors show $(\text{EWDE}_{\text{NHST}} - \text{EWDE}_{\text{eJAB}})/M$, where each panel reports its maximum absolute EWDE difference M over the displayed grid. Positive values (yellow) favor eJAB; negative values (purple) favor NHST. Dashed curves represent local alternatives of the form $\theta = c/\sqrt{n}$. The corresponding $p \leq .005$ comparison appears in Panel A of manuscript Figure 2.

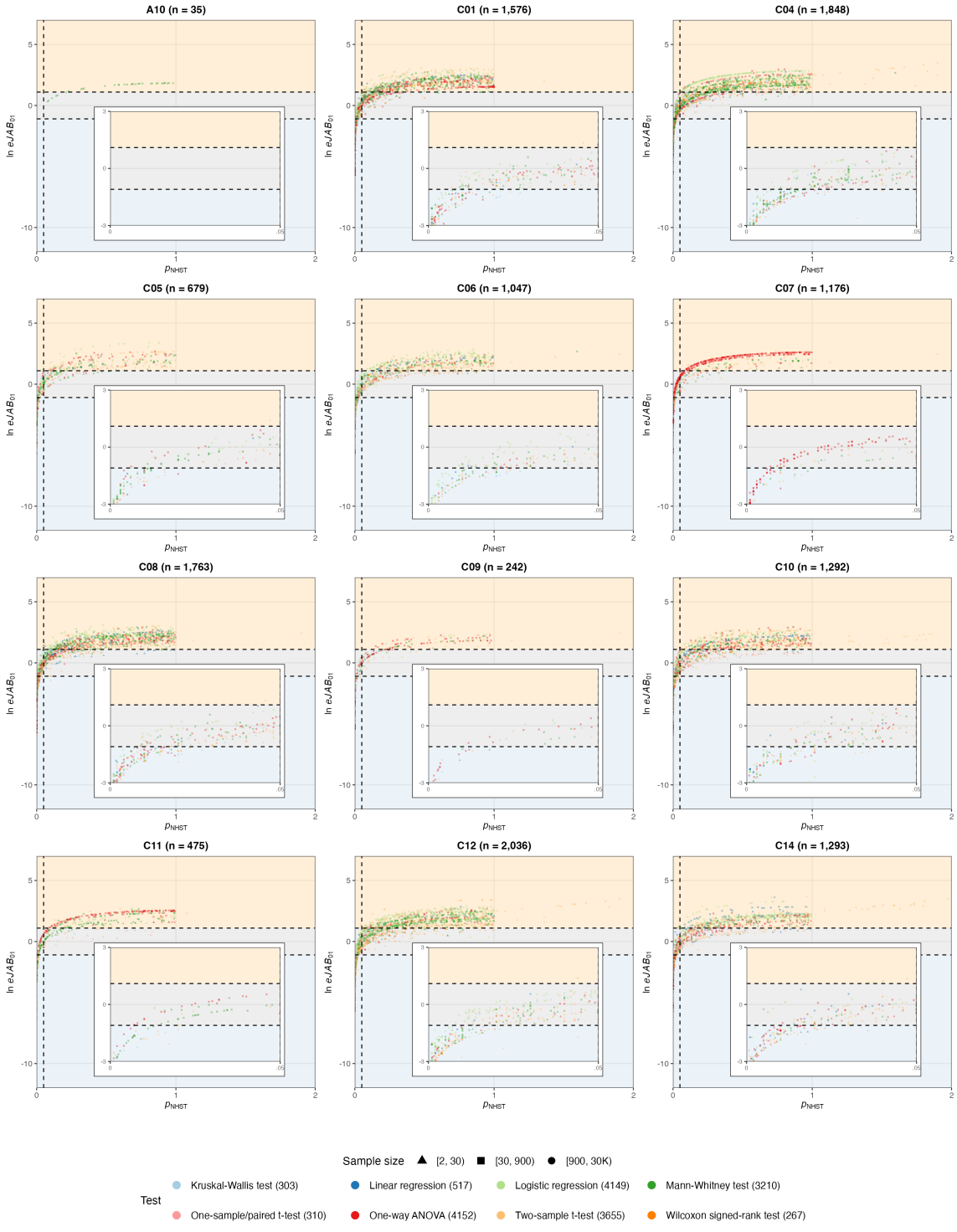


Figure S4: Scatterplots of ln $eJAB_{01}$ versus p_{NHST} by MeSH condition (part 1 of 3), with $p_{\text{NHST}} = p$ for two-sided tests and $2p_1$ for one-sided tests. Insets show $p_{\text{NHST}} \leq .05$. Colors identify tests, shapes sample-size groups, and shading evidence categories.

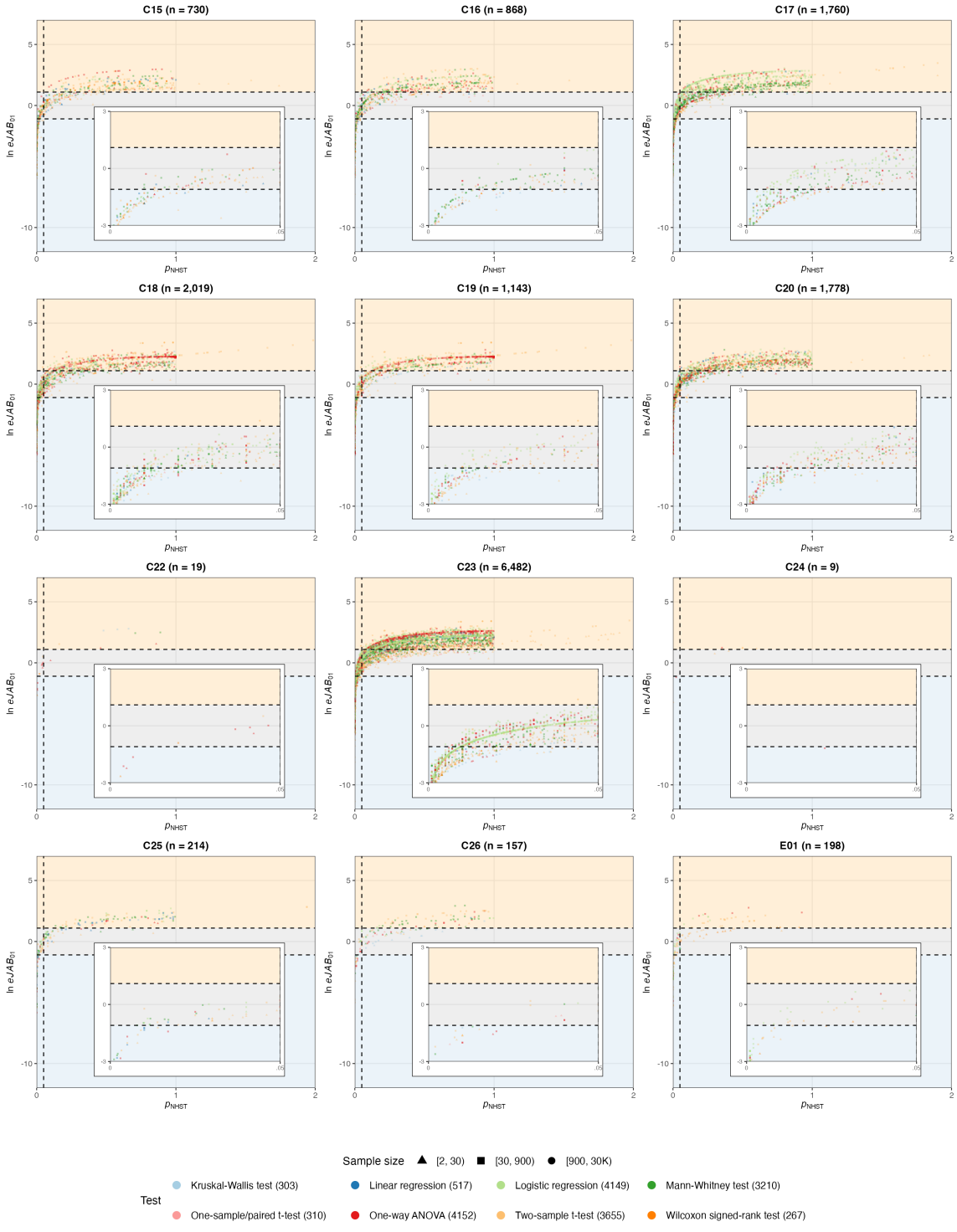


Figure S5: Scatterplots of ln $eJAB_{01}$ versus p_{NHST} by MeSH condition (part 2 of 3), with $p_{\text{NHST}} = p$ for two-sided tests and $2p_1$ for one-sided tests. Insets show $p_{\text{NHST}} \leq .05$. Colors identify tests, shapes sample-size groups, and shading evidence categories.

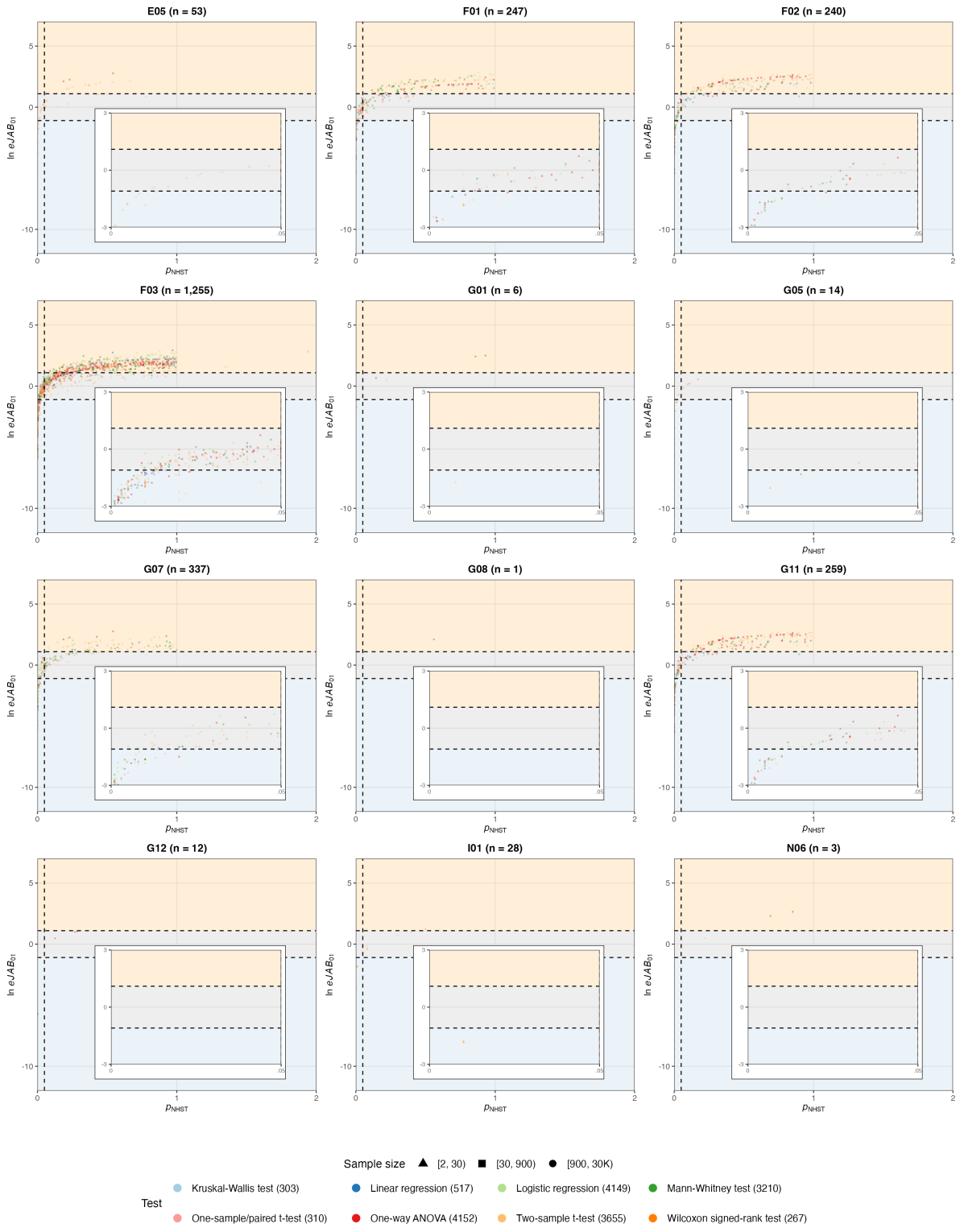


Figure S6: Scatterplots of $\ln eJAB_{01}$ versus p_{NHST} by MeSH condition (part 3 of 3), with $p_{\text{NHST}} = p$ for two-sided tests and $2p_1$ for one-sided tests. Insets show $p_{\text{NHST}} \leq .05$. Colors identify tests, shapes sample-size groups, and shading evidence categories.

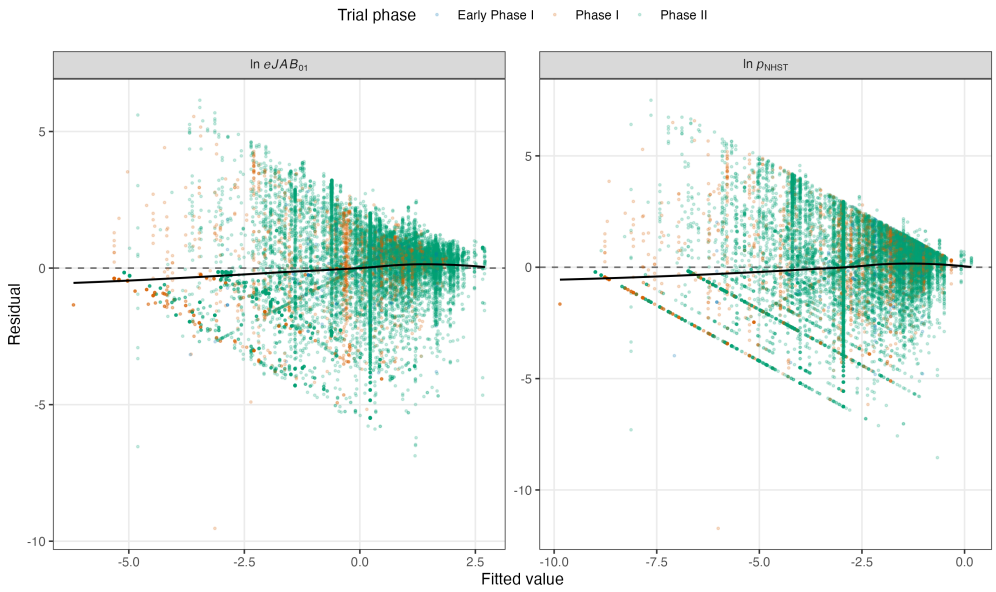


Figure S7: Residuals versus fitted values from random-intercept models for $\ln eJAB_{01}$ (left) and $\ln p_{\mathrm{NHST}}$ (right), where $p_{\mathrm{NHST}} = p$ for two-sided tests and $2p_1$ for one-sided tests, in the screened primary- and secondary-outcome set. The dashed horizontal line marks zero residuals and the solid curve is a LOESS smooth. Nearly parallel slanted bands reflect heaping at common reported p thresholds.

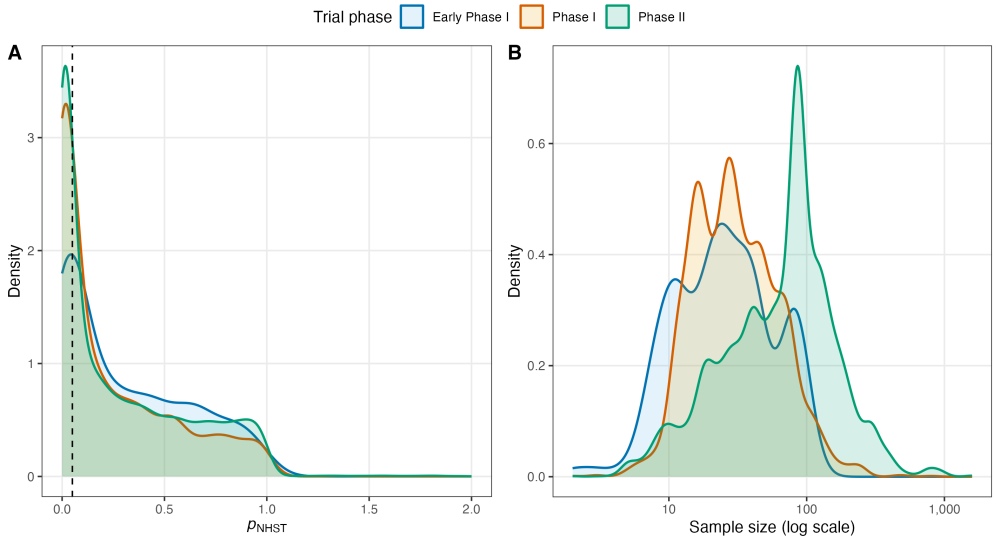


Figure S8: Distributions of p_{NHST} (A; vertical dashed line at .05), equal to p for two-sided tests and $2p_1$ for one-sided tests, and sample size (B) by exact Early Phase I, Phase I, and Phase II classification in the screened primary- and secondary-outcome set.

Appendix C: Detailed ClinicalTrials.gov Inclusion and Exclusion Criteria

C.1 Scope, archived source, and units

The source was the ClinicalTrials.gov (CTG) JSON results export for completed studies with posted results, downloaded on August 7, 2025. The extraction emitted one row for every participant-denominator group attached to an outcome–analysis pair. A deterministic MD5 hash of the CTG identifier, the complete outcome object, and the complete analysis object (with sorted JSON keys) defined `analysisId`. Thus, `analysisId`, rather than an individual denominator row or a CTG study, was the unit of statistical analysis. Rows sharing an `analysisId` were group-denominator components of the same test and were collapsed during sample-size construction. Statistical analyses were nested within CTG studies identified by `nctId`.

The manuscript analysis estimates the frequency of disagreement between NHST significance and eJAB evidence among eligible primary- and secondary-outcome analyses from exact Early Phase I, exact Phase I, and exact Phase II interventional trials. Both one-sided and two-sided/unrestricted tests are included, using $p_1 \leq .025$ and $p \leq .05$, respectively. The three phase classifications are retained as separate groups. These criteria define the screened set used throughout the manuscript and supplement.

C.2 Inclusion criteria

An analysis was eligible only when all of the following criteria were met:

1. The record occurred in the archived CTG results extraction and had a deterministic `analysisId` and `nctId`.
2. CTG classified the study type as `INTERVENTIONAL`.
3. CTG classified the study as *exactly* Early Phase I, *exactly* Phase I, or *exactly* Phase II. Exact Early Phase I required `EARLY_PHASE1`=1 and `PHASE1`=`PHASE2`=`PHASE3`=`PHASE4`=0. Exact Phase I and exact Phase II were defined analogously using their respective indicator.
4. CTG labeled the outcome as `PRIMARY` or `SECONDARY`.
5. The reported p-value contained a numeric value strictly between 0 and 1 and was not right-censored (for example, $p > .05$).
6. The free-text statistical method mapped deterministically to one of the eJAB-computable classes in Table S2.
7. All participant denominators required for the mapped test were finite and positive; the derived sample size satisfied $n > 1$; and the test dimension satisfied $q \geq 1$.
8. The resulting direction-appropriate eJAB was finite and positive.

Statistical significance was *not* an eligibility criterion. Both significant and nonsignificant eligible results were retained for the descriptive analysis and trial-cluster resampling.

C.3 Exclusion criteria

The following were excluded, in the sequential order shown in Table S1:

- Four prespecified malformed analysis identifiers previously found to contain denominator errors.
- Observational and expanded-access records, and any record not labeled interventional.
- Phase I/Phase II, Phase II/Phase III, Phase III, Phase IV, not-applicable, or missing phase classifications. Combined phases were not recoded into any of the three early-phase groups.
- Outcomes other than primary or secondary, baseline measures, participant-flow analyses, and adverse-event analyses.

- Missing, blank, nonnumeric, zero, one, or otherwise invalid p-values. Right-censored values were excluded because their numerical bound does not identify the value required for eJAB. Left-censored values were handled as described in Section C.5.
- Methods outside the prespecified mappings, including ANCOVA, mixed-effects or repeated-measures models, Fisher exact tests, chi-squared tests without full table dimensions, Cox or log-rank analyses without the number of observed events, Bayesian analyses, and other free-text methods that could not be mapped without assumptions not supported by the CTG record.
- Analyses with missing, zero, or ambiguous participant denominators; one-sample or signed-rank analyses represented by more than one denominator row; omnibus analyses represented by fewer than two groups; $n \leq 1$; or $q < 1$.
- A nonfinite or nonpositive eJAB value after all preceding steps.

Table S1: Sequential inclusion and exclusion flow. Counts are analysis identifiers except in the last column. Each exclusion count is conditional on all previous steps.

Screening stage	Excluded	Retained	Studies
Unique analyses in the archived extraction	0	279,927	22,454
Remove four malformed analysis identifiers	4	279,923	22,453
Restrict to interventional studies	11,893	268,030	21,500
Restrict to exact Early Phase I, Phase I, or Phase II	175,191	92,839	6,302
Restrict to primary and secondary outcomes	1,806	91,033	6,277
Require an admissible p-value and exclude right-censoring	26,665	64,368	4,461
Require a prespecified eJAB method mapping	47,214	17,154	1,877
Require valid denominators, row structure, n, and q	591	16,563	1,790

C.4 Method mapping, dimension, and sample size

Free-text method labels were normalized using case-insensitive regular expressions. More specific patterns were evaluated before broader patterns (for example, conditional logistic regression before logistic regression, and one-sample or paired t-tests before other t-tests). Table S2 gives the complete set used for the early-phase reanalysis.

Table S2: Eligible statistical method classes and derived quantities. Here G is the number of attached denominator groups.

Mapped class	Sample-size rule	Dimension q
One-sample or paired t-test	One attached denominator; n is that value	1
Two-sample t-test	Sum attached group denominators	1
Linear regression	Sum attached denominators	1
Logistic regression	Sum attached denominators	1
Conditional logistic regression	Maximum attached denominator	1
One-way ANOVA	Sum attached group denominators; require $G \geq 2$	$G - 1$
Wilcoxon signed-rank test	One attached denominator; n is that value	1
Mann–Whitney test	Sum attached group denominators	1
Kruskal–Wallis test	Sum attached group denominators; require $G \geq 2$	$G - 1$

Participant counts containing thousands separators were parsed as numbers before aggregation. A missing or nonpositive attached denominator made the full analysis ineligible; missing values were not silently removed from a sum. The archived extraction's repeated `analysisId` rows encode attached denominator groups and were therefore aggregated, not counted as duplicate hypothesis tests. The hash construction produced one final record per outcome–analysis pair; no additional duplicate analysis identifiers remained after aggregation.

C.5 Reported inequalities, test direction, and Wagenmakers's one-sided Bayes factor

The numeric component of a reported p-value was extracted reproducibly. For a left-censored value such as $p < .05$, the bound .05 was used in the eJAB calculation. This is conservative for evidence against the null because the unknown underlying value is smaller. The reported inequality itself was retained: a report of $p < .05$ met the $p \leq .05$ decision rule even though the stored numerical bound equaled .05. Numerical reports were compared as displayed, so a reported $p = .05$ was also included (and likewise $p_1 = .025$ for a one-sided test). A rounded value at the threshold does not establish that the underlying full-precision value satisfies the rule; these classifications describe the reported values. Values reported with ">" were excluded because a lower bound does not identify the tail probability needed for eJAB. The final eligible set contained 2,714 left-censored values (30 in Early Phase I, 429 in Phase I, and 2,255 in Phase II).

Direction was determined from the reported `statisticalMethod`. The patterns `one-sided`, `1-sided`, `one-tailed`, and `1-tailed` identified a one-sided test. A directional correction was used only for scalar tests ($q = 1$). An explicit two-sided label or the absence of an explicit one-sided method label was classified as two-sided/unrestricted. Mentions of one-sided α found only in sample-size or power prose were not allowed to reclassify the reported test. This rule identified 427 one-sided analyses and 16,136 two-sided/unrestricted analyses.

For a one-sided reported value p_1, the corresponding two-sided value was first reconstructed as

$$p = 2\min\{p_1, 1 - p_1\}.$$

For two-sided/unrestricted analyses, eJAB was computed as

$$eJAB_{01} = \sqrt{n}\exp\left\{-\frac{1}{2}\left(1 - n^{-1/q}\right) Q_{\chi^2_q}(1-p)\right\}.$$

For one-sided scalar analyses, we used the null-centered definition in the manuscript, combining the Savage–Dickey approximation with the directional correction of Wagenmakers (2022, pp. 20–22):

$$eJAB_{0+} = \frac{\sqrt{n}}{2(1-p_1)}\exp\left\{-\frac{1}{2}Q_{\chi^2_1}(1-p)\right\}.$$

This null-centered construction omits the finite-sample factor $(n-1)/n$ from the exponent. The directional correction uses prior mass 1/2 and approximate posterior mass $1 - p_1$ in the predicted direction for moderate samples and a relatively vague local prior. Thus the procedure does not merely substitute a one-sided p-value into a chi-squared formula.

Statistical significance was defined as $p_1 \leq .025$ for one-sided analyses and $p \leq .05$ for two-sided/unrestricted analyses. All clinical comparisons and plots use $p_{\mathrm{NHST}} = 2p_1$ or $p_{\mathrm{NHST}} = p$, respectively. At each sweep value α, a *Bayes/NHST contradiction* requires $p_{\mathrm{NHST}} \leq \alpha$ and the applicable $eJAB_{01}$ or $eJAB_{0+}$ exceeding 1. Thus the one-sided condition is $p_1 \leq \alpha/2$, preserving the specified direction and agreeing with the reported counts at $\alpha = .05$. Values of p_{NHST} above one are retained as scaled comparison values, not interpreted as probabilities. The folded value $2\min\{p_1, 1 - p_1\}$ is used only inside eJAB: using it for NHST would incorrectly admit opposite-tail results. This descriptive term does not assert that the result is a type I error.

C.6 Final analytic set and reproducibility

The final set comprised 16,563 primary- and secondary-outcome analyses from 1,790 trials: 270 analyses from 47 exact Early Phase I trials, 2,098 analyses from 280 exact Phase I trials, and 14,195 analyses from 1,463 exact Phase II trials. Combining one-sided tests with $p_1 \leq .025$ and two-sided/unrestricted tests with $p \leq .05$, 94 Early Phase I, 927 Phase I, and 5,453 Phase II analyses were NHST significant. Six Early Phase I results (6.38% of all significant results in that phase), 28 Phase I results (3.02%), and 459 Phase II results (8.42%) were Bayes/NHST contradictions, with the applicable eJAB > 1. The significant denominators include 4, 27, and 104 one-sided results, respectively; none has one-sided eJAB > 1.

The complete executable pipeline is `analysis/early_phase/early_phase_analysis.R` in the GitHub repository. It writes the sequential flow table, the eligible analysis-level data set, phase and test summaries, and the clustered mixed-model sensitivity result. The four manual exclusions, method regular expressions, direction rules, significance thresholds, denominator rules, and eJAB calculations are encoded directly in that script. The supplementary JLP catalogue is generated from this eligible set by `analysis/early_phase/jlp_catalog.R`; its machine-readable output retains the complete analysis identifiers and archived source descriptions.

The July 2026 protocol was archived after the August 2025 CTG snapshot had already been downloaded and inspected. Accordingly, this is described as a protocol-aligned retrospective reanalysis, not as a prospective confirmatory study or an independent replication. For phase g, let N_g count all eligible analysis records, S_{ig} the significant records in trial i at $p_{\mathrm{NHST}} \leq .05$, and $R_{ig}(K)$ those significant records with applicable eJAB $> K$. Define $S_g = \sum_i S_{ig}$ and $R_g(K) = \sum_i R_{ig}(K)$. Manuscript Figure 3A plots the overall proportion $R_g(K)/N_g$ as the BF threshold K varies, marking contradictions at $K = 1$ and the stronger Jeffreys–Lindley screen at $K = 3$. Figure 3B displays bootstrap distributions of the conditional proportions

$$\widehat{r}_g(K) = \frac{R_g(K)}{S_g} = \frac{\sum_i R_{ig}(K)}{\sum_i S_{ig}}, \qquad K \in \{1, 3\}.$$

Both summaries weight analysis records equally within phase, rather than averaging trial-specific proportions or estimating the fraction of trials containing a flag. For $K = 3$, the conditional proportions are 0/94 (0%), 2/927 (0.22%), and 18/5,453 (0.33%); the corresponding overall proportions are 0/270 (0%), 2/2,098 (0.095%), and 18/14,195 (0.127%). The 20 screened records include the reporting and hypothesis qualifications described in Section C.7, and these are screening proportions rather than rates of confirmed paradox instances. In particular, both Phase I records are bound-only reports; Phase II contains the 11 numerical reports and seven bound-only reports. Supplementary Figure S11 retains the complementary NHST significance-level sweep at the fixed BF threshold 1. Manuscript Figure 3C restores the scatterplot display to $0 \leq p_{\mathrm{NHST}} \leq 1$; the 74 records with $p_{\mathrm{NHST}} > 1$ remain in the eligible set and all summary calculations.

The trial-cluster bootstrap resamples eligible trials uniformly with replacement within each phase, keeping all records from each selected trial together, and recomputes each ratio of totals. Trials with no significant records remain in the resampling frame and contribute zero to both sums. The 95% limits are the 2.5th and 97.5th percentiles of 20,000 replicates; a replicate with a zero significant denominator is undefined, not zero. These distributions describe resampling variability under an exchangeability assumption for comparable eligible trials; they are not posterior distributions or estimates of the probability that the null hypothesis is true. They do not account for selective reporting, ambiguous hypotheses, p-value censoring, or approximation error. Early Phase I has only five trials contributing contradictions, so its interval requires particular caution. The previous study-level Dirichlet analysis is retained only as the audit and prior-sensitivity analysis in Section C.8.

Figure 3B overlays separate phase-specific kernel density estimates for $K = 1$ and $K = 3$, with dashed vertical lines at the observed conditional proportions. Each bootstrap distribution is represented as a

point mass at zero plus a continuous component. Let $\widehat{w}_g$ be the fraction of valid bootstrap replicates equal to zero. A Gaussian kernel density is fitted only to positive replicates, using the `stats::bw.nrd0` bandwidth for that phase and threshold; reflection at zero prevents smoothing onto negative proportions. The reflected density is multiplied by $1 - \widehat{w}_g$, and $\widehat{w}_g$ is displayed separately as a probability mass, not a density height. For Early Phase I at $K = 3$, every bootstrap replicate is zero, so there is no continuous density. The empirical bootstrap cannot generate a screened event absent from every sampled trial; this degeneracy does not establish zero population prevalence or quantify uncertainty about unobserved events.

Using seed 20261105 for $K = 1$, the intervals are 1.23–13.75% for Early Phase I, 1.14–5.86% for Phase I, and 6.62–10.12% for Phase II; no bootstrap replicate has a zero significant denominator. These draws are preserved from `analysis/early_phase/clinical_phase_summary.R`. The $K = 3$ resamples use seed 20261106 in `analysis/early_phase/jlp_phase_panels.R`, which also writes the threshold profiles and density summaries. Their percentile intervals are 0–0% for Early Phase I, 0–0.575% for Phase I, and 0.086–0.688% for Phase II; no replicate has a zero significant denominator. The degenerate Early Phase I interval has the limitation described above. The observed bootstrap masses at zero are 0.52%, 0%, and 0% at $K = 1$, and 100%, 13.845%, and 0.105% at $K = 3$, in the same phase order. As an influence check for $K = 1$, omitting one trial at a time gives observed-proportion ranges of 4.35–7.06%, 1.97–3.19%, and 7.80–8.75%, respectively. These ranges are sensitivity diagnostics, not uncertainty intervals. The largest Phase II cluster contributes 646 of its 5,453 significant records, illustrating why records from one trial should be kept together when resampling.

C.7 Catalogue of screened Jeffreys–Lindley cases

We screened the eligible set for reported NHST significance, $p_{\mathrm{NHST}} \leq .05$, together with the applicable eJAB > 3. This stronger disagreement threshold is a descriptive criterion for identifying finite-sample examples compatible with the Jeffreys–Lindley phenomenon under the stated eJAB approximation; it does not establish that a scientific effect is absent. The screen returns 20 analysis records from nine trials: two in Phase I and 18 in Phase II, with none in Early Phase I. All 20 were mapped to two-sided/unrestricted scalar tests ($q = 1$) by the method-label rule.

The catalogue distinguishes numerical reports from bounds. Eleven records, all in Phase II and drawn from five trials, have numerical reported p-values and yield point-calculated eJAB values above 3. The other nine report only $p < .05$: evaluating eJAB at .05 gives an *upper bound* on the null evidence, so eJAB > 3 cannot be established for those records. Four of these nine also have equivalence or bioequivalence context in the archived analysis descriptions; the generic method-label mapping does not justify interpreting their p-values as ordinary tests of the point null used by eJAB. They are retained in the screening catalogue for transparency and are not presented as confirmed JLP examples. Numerical reports may themselves be rounded, and the classification refers to the displayed values, not unavailable full-precision values. The recorded method labels and available archived descriptions also do not independently verify every hypothesis specification or complete-case sample size.

Table S3 gives each reported finding and comparison, its method, participant-based sample size, reported p-value, eJAB value or bound, and qualification. Separate outcome–analysis records from one trial are retained as separate rows and are not independent trial replications. Linked registry identifiers allow inspection of the trial; the descriptions and calculations here use the archived August 7, 2025 extract, which may differ from the current registry record.

Table S3: Early-phase records with reported NHST significance and computed $eJAB_{01} > 3$. The screening set includes literal p-value bounds; flagged rows require the qualifications below.

Trial / phase	Outcome or comparison	Method / n	Reported p	$eJAB_{01}$	Flag
J01 NCT00071513 Phase I	Change in Short Moods and Feelings Questionnaire (SMFQ)	Two-sample t-test $n = 480$	< 0.05	≤ 3.23	B
J02 NCT02260492 Phase I	FEV1 Trough	ANOVA $n = 801$	< 0.05	≤ 4.16	B,E
J03 NCT00405821 Phase II	HIV-1 Viral Load Difference Between Arms	Two-sample t-test $n = 440$	0.05	3.09	N
J04 NCT00565812 Phase II	Number of Participants With Decrease in Total Analgesic Medication Use	Logistic regression $n = 971$	0.032	3.13	N
J05 NCT00565812 Phase II	Number of Participants With Joint Space Narrowing Progression	Logistic regression $n = 848$	0.047	4.06	N
J06 NCT01225822 Phase II	Major bleeding events: BIBR 1048 50 mg twice daily versus enoxaparin 40 mg daily	Logistic regression $n = 781$	0.0472	3.91	N
J07 NCT01225822 Phase II	Proximal DVT: BIBR 1048 225 mg versus 50 mg, both twice daily	Logistic regression $n = 753$	0.036	3.05	N

Trial / phase	Outcome or comparison	Method / n	Reported p	$eJAB_{01}$	Flag
J08 NCT01225822 Phase II	Proximal DVT, pulmonary embolism and VTE-related mortality: BIBR 1048 225 mg versus 50 mg, both twice daily	Logistic regression $n = 753$	0.0373	3.15	N
J09 NCT01225822 Phase II	VTE events and all-cause mortality: BIBR 1048 150 mg twice daily versus enoxaparin 40 mg daily	Logistic regression $n = 757$	0.0401	3.36	N
J10 NCT01225822 Phase II	VTE events: BIBR 1048 150 mg twice daily versus enoxaparin 40 mg daily	Logistic regression $n = 757$	0.0401	3.36	N
J11 NCT01434680 Phase II	Meningococcal C hSBA geometric mean titers at day 28: MenC-CRM ROS versus EMV	ANOVA $n = 540$	< 0.05	≤ 3.42	B,E
J12 NCT01434680 Phase II	Meningococcal C hSBA geometric mean titers at day 28: MenC-CRM LIQ versus EMV	ANOVA $n = 564$	< 0.05	≤ 3.50	B,E
J13 NCT01434680 Phase II	Meningococcal C hSBA geometric mean titers at day 28: MenC-CRM liquid versus ROS	ANOVA $n = 542$	< 0.05	≤ 3.43	B,E
J14 NCT02303704 Phase II	Pharamacological Inotropic Support (Dobutamine)	Two-sample t-test $n = 434$	< 0.05	≤ 3.07	B
J15 NCT02303704 Phase II	Pharmacologic Inotropic Support (Adrenaline)	Two-sample t-test $n = 434$	< 0.05	≤ 3.07	B
J16 NCT02303704 Phase II	Pharmacological Inotropic Support (Nor-adrenaline)	Two-sample t-test $n = 434$	< 0.05	≤ 3.07	B
J17 NCT02303704 Phase II	Post-op CK-MB Levels	Two-sample t-test $n = 434$	< 0.05	≤ 3.07	B
J18 NCT04376684 Phase II	Part 1: Percentage of Participants Alive and Free of Respiratory Failure at Day 28	Logistic regression $n = 782$	0.0456	3.80	N
J19 NCT05013008 Phase II	Plasma biomarker change: CCN family member 4 (O95388)	Two-sample t-test $n = 921$	$= 0.044$	4.00	N
J20 NCT05013008 Phase II	Plasma biomarker change: TGF beta-1 proprotein (P01137)	Two-sample t-test $n = 921$	$= 0.044$	4.00	N

N: numeric p report; B: left-censored p report. For B rows, the displayed eJAB is evaluated at the reported upper bound and is itself an upper bound (rounded upward for display), so $eJAB_{01} > 3$ is not established. E: archived descriptions indicate equivalence or bioequivalence hypotheses; an ordinary point-null interpretation is not justified without further review. NCT02260492 refers to a Day 1 analysis that explicitly describes two one-sided bioequivalence tests. All listed analyses were classified as two-sided/unrestricted in the screening pipeline and have $q = 1$; significance includes $p = .05$. Participant counts are sums of the distinct outcome-level groups associated with each analysis; they do not independently verify the analysis-specific complete-case sample size. Multiple outcomes or biomarkers within one trial are separate analyses, not independent trials. Numeric reports remain subject to rounding and the approximation's assumptions. Row identifiers J01–J20 map to the complete archived analysis identifiers in `jlp_cases.csv`.

C.8 Audit of empirical and posterior contradiction proportions

We reproduced the original submission and the revised analysis at both eJAB thresholds, using the same significance rule for the empirical and posterior quantities within each comparison. Table S4 uses the inclusive NHST rule throughout: $p \leq .05$ in the original cohort, and $p \leq .05$ or $p_1 \leq .025$ as applicable in the revised cohort. Both current revision quantities used eJAB > 1; their discrepancy was not caused by a 1/3 versus 1 threshold mismatch. The original Figure 3E, however, was computed with eJAB $> 1/3$ and $p < .05$, whereas the original contradiction count used eJAB > 1 and $p \leq .05$. Re-executing the original function reproduces its posterior mean of 35.52%, with the matching empirical count 8,088/27,505 (29.41%). Neither data set contains an eJAB value exactly equal to 1 or 1/3, so changing strictness at those BF boundaries does not alter these counts.

Table S4: Matched empirical and previous-model posterior summaries. Posterior means use a separate Dirichlet$(1, 1, 1, 1)$ prior for every study; they are shown for audit, not used as the main estimates.

Cohort	eJAB threshold	Count	Observed (%)	Posterior mean (%)
Original	$> 1/3$	11,369/30,790	36.92	39.82
Original	> 1	4,088/30,790	13.28	22.77
Revised	$> 1/3$	1,994/6,474	30.80	34.99
Revised	> 1	493/6,474	7.62	17.95

The original cohort contains 71,126 results from 9,002 studies, compared with 16,563 results from 1,790 trials in the revision. Restricting the original cohort to the revised phase, study-type, and outcome scope reduces the observed eJAB > 1 proportion from 13.28% to 544/6,955 (7.82%); the remaining preprocessing changes yield 7.62%. Thus the change in the raw proportions and the displacement of a posterior from its own raw proportion are distinct issues.

The previous model assigned an independent Dirichlet(a, a, a, a) prior to each trial's four categories defined by NHST significance and whether eJAB exceeded the chosen threshold. It then weighted each trial's posterior category probabilities by its number n_i of eligible analyses before conditioning on significance. It was not a single update of the pooled phase counts or a hierarchical model estimating a shared distribution. Writing s_i and c_i for the significant and contradictory counts, the ratio of the posterior expected weighted masses is

$$\frac{\sum_i n_i(c_i + a)/(n_i + 4a)}{\sum_i n_i(s_i + 2a)/(n_i + 4a)}.$$

This diagnostic ratio is not generally the posterior expectation of the ratio, but is numerically close here. For Phase II at $a = 1$, its numerator separates into 359.06 from observed counts and 689.42 from prior terms; the denominator separates into 4,408.88 and 1,378.85. Their ratio is 18.12%, compared with the observed 459/5,453 (8.42%). There are 1,463 separate priors, the median trial contributes three eligible and one significant record, and 580 trials have no observed significant record. A large pooled denominator therefore does not make this repeated prior contribution negligible.

Table S5: Sensitivity of the previous posterior mean for eJAB > 1 to the prior parameter a in each of four categories per trial. Values are percentages. Smaller a values diagnose prior influence; they were not selected as replacement priors.

Phase	Observed	$a = 1$	$a = 0.1$	$a = 0.01$
Early Phase I	6.38	24.15	9.67	6.76
Phase I	3.02	16.27	5.35	3.28
Phase II	8.42	18.12	10.26	8.63

The substantial sensitivity in Table S5 precludes using the previous posterior phase differences as conclusions primarily supported by the observed counts. The main analysis instead reports the empirical proportions with trial-cluster bootstrap intervals (Section C.6); Figure S10 preserves the previous density comparison as an audit. The same correction applies to the less stringent eJAB $> 1/3$ summary: its observed revised proportion is 1,994/6,474 (30.80%), while 34.99% is the mean under the previous model. Executable reconstruction, threshold checks, and independent prior-decomposition calculations are retained under `analysis/audits/posterior_20261104/` in the repository.

Appendix D: Proof of Model-Selection Consistency

Lemma 1. *For $x \to 1^-$,*

$$0 < f(x) = -2\ln(1-x) - 2\ln(-\ln(1-x)) < Q_{\chi^2_q}(x).$$

The proof is provided in the GitHub repository (see files within 'Theorem').

PROOF OF THEOREM 2.3. Assume $\mathcal{H}_0$ is true. Let $\delta \in (0,1)$ and $K > 0$ be arbitrary. Note that **R1** and $W_n = Q_{\chi^2_q}(1-p) \xrightarrow{D} \chi^2_q$ by the continuous mapping theorem.
For $n \geqslant N^* = \lceil K^2 \exp\{Q_{\chi^2_q}(1-\delta)\} \rceil$, we have

$$\begin{aligned}
&\mathbb{P}(eJAB_{01} > K) = \mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\frac{n^{1/q}-1}{n^{1/q}}\,W_n\right\} > K\right)\\
>&\mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\,W_n\right\} > K\right)\\
=&\mathbb{P}\left(W_n < -2\ln\frac{K}{\sqrt{n}}\right)\\
\geqslant&\mathbb{P}\left(W_n < -2\ln\frac{K}{\sqrt{N^*}}\right)\\
\geqslant&\mathbb{P}\left(W_n < -2\ln\frac{K}{\sqrt{K^2\exp\{Q_{\chi^2_q}(1-\delta)\}}}\right)\\
=&\mathbb{P}\Big(W_n < Q_{\chi^2_q}(1-\delta)\Big) \to 1-\delta \text{ as } n \to \infty.
\end{aligned}$$

Thus, $eJAB_{01} \xrightarrow{P} \infty$ as $n \to \infty$ under $\mathcal{H}_0$.

Assume $\mathcal{H}_1$ is true. Let $\delta \in (0,1)$ and $\epsilon > 0$ be arbitrary. Lemma 1 gives $0 < f(1-p) < Q_{\chi^2_q}(1-p)$ for $p \to 0$.

$$\begin{aligned}
&\mathbb{P}(eJAB_{01} < \epsilon)\\
=&\mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\frac{n^{1/q}-1}{n^{1/q}}\,Q_{\chi^2_q}(1-p)\right\} < \epsilon\right)\\
>&\mathbb{P}\left(\sqrt{n}\exp\left\{-\frac{1}{2}\frac{n^{1/q}-1}{n^{1/q}} f(1-p)\right\} < \epsilon\right)\\
=&\mathbb{P}\left(\sqrt{n}\ \left(p\cdot(-\ln p)\right)^{\frac{n^{1/q}-1}{n^{1/q}}} < \epsilon\right) > 1-\delta.
\end{aligned}$$

The final inequality is true for n sufficiently large by **R2** and the continuous mapping theorem with $1 - \frac{1}{n^{1/q}} \to 1$.

Thus, $eJAB_{01} \xrightarrow{P} 0$ as $n \to \infty$ under $\mathcal{H}_1$. Together, *eJAB* is model-selection consistent.

□

Appendix E: Realized Values of D_n in Four Replication Projects

Considering **R2** of Theorem 2.3 from a practical standpoint, we examine the realized values of $D_n = -\sqrt{n} \cdot p \ln p$ in four large-scale replication projects. Each project selected a collection of previously published findings and attempted to replicate them under a preregistered protocol, so that each original result was paired with an independently conducted replication of the same effect.

- **EERP**, the Experimental Economics Replication Project (Camerer et al., 2016, *Science*), which attempted to replicate laboratory experiments published in the *American Economic Review* and the *Quarterly Journal of Economics* between 2011 and 2014;
- **RPP**, the Reproducibility Project: Psychology (Open Science Collaboration, 2015, *Science*), which attempted to replicate a collection of findings drawn from three high-impact psychology journals;
- **RPCB**, the Reproducibility Project: Cancer Biology (Errington et al., 2014, *eLife*), which considered experiments reported in high-impact preclinical cancer biology papers; we use the project's final analysis (Errington et al., 2021, *eLife*); and
- **SCORE**, Systematizing Confidence in Open Research and Evidence (Tyner et al., 2025, *Nature*), which assembled and considered replication of a large corpus of published claims from the social and behavioral sciences.

For each result in the four projects, we extracted the original p-value and sample size n, and the p-value and sample size of each of its replication attempts. Each result has at least one and possibly several replication attempts. A result is included in the $\mathcal{H}_1$ subset when its original p-value satisfies $p \leq 0.005$ and every one of its replication attempts also reached $p \leq 0.005$. Given that the effect was found at this level in the original study and in every replication, we make the assumption that $\mathcal{H}_1$ is true for these results. This gives realized values of D_n under $\mathcal{H}_1$: one from each original study and one from each replication attempt. Table S6 reports the counts at each step.

Two kinds of recorded p-value were treated separately. Where a p-value was recorded as 0, we computed $\ln p$ from the reported test statistic and, where the test has them, its degrees of freedom. Of the plotted values, 51 were obtained this way, including the value at the largest sample size; 12 values (six SCORE original results and six SCORE replication attempts) could not be computed and were dropped. Where a p-value was reported as a bound, such as $p < .001$, rather than as a value, we removed it. Such bounds occur only in SCORE. SCORE labels its original p-values; we removed the 37 it labels as less-than or threshold values, and the one with no label. SCORE does not label replication p-values, so we treated a replication p-value recorded as an exact power of ten, such as .001, as censored unless we were able to recompute the p-value directly from the reported test statistic. This removed two replication values where the p-value is left censored and cannot be recalculated from the reported test statistic.

The columns of Table S6 are as follows.

(1) *Released*: the number of results in each project's released data. A result is a study for EERP (18) and RPP (167, of which 100 were replicated), an effect for RPCB (158 effects from 23 papers, given in 188 rows), and a claim for SCORE (331 claims with at least one replication report).
(2) *With p and n*: the results in column 1 for which both the original p-value and the original sample size are recorded.
(3) *Original $p \leq 0.005$*: the results in column 2 whose original p-value satisfies $p \leq 0.005$.
(4) *$\mathcal{H}_1$ subset*: the results in column 3 for which every replication attempt reached $p \leq 0.005$. A result with no recorded replication p-value is not included; this excludes 20 RPP results and one RPCB result. We also excluded RPP study 69. The RPP data record $p = 10^{-6}$ for both its original and its replication, with no test statistic from which either value could be checked, whereas the reported p-values in the same data are $p < .05$ for the original and $p = .05$ for the replication. For SCORE, a replication attempt is one replication report.
(5) *Plotted values*: the values of D_n shown in Figure S9, one for each original result and one for each

replication attempt in the $\mathcal{H}_1$ subset, less the p-values reported as bounds and the zero p-values whose $\ln p$ could not be computed. For RPCB and SCORE the replication column exceeds the $\mathcal{H}_1$ subset because a result can have more than one replication attempt (up to three in RPCB and four in SCORE), each contributing its own value.

Project	Field	Results				(5) Plotted values	
		(1) released	(2) with p, n	(3) $p \leq 0.005$	(4) $\mathcal{H}_1$ subset	original	replication
EERP	Experimental economics	18	18	7	5	5	5
RPP	Psychology	167	143	67	16	16	16
RPCB	Preclinical cancer biology	158	115	61	15	15	21
SCORE	Social and behavioral sciences	331	311	184	102	58	107
	Total	674	587	319	138	94	149

Table S6: Counts at each step of the selection, by project. The columns are described in the notes above the table.

Figure S9 plots the realized values of D_n against the sample size n, which spans roughly seven orders of magnitude. The most important of the realized values is the one at the largest sample size, which is circled in the figure. Based on this, the condition **R2** seems feasible. The data, code and counts for Table S6 and Figure S9 are in the GitHub repository (see 'Supplementary Information/R2Empirical').

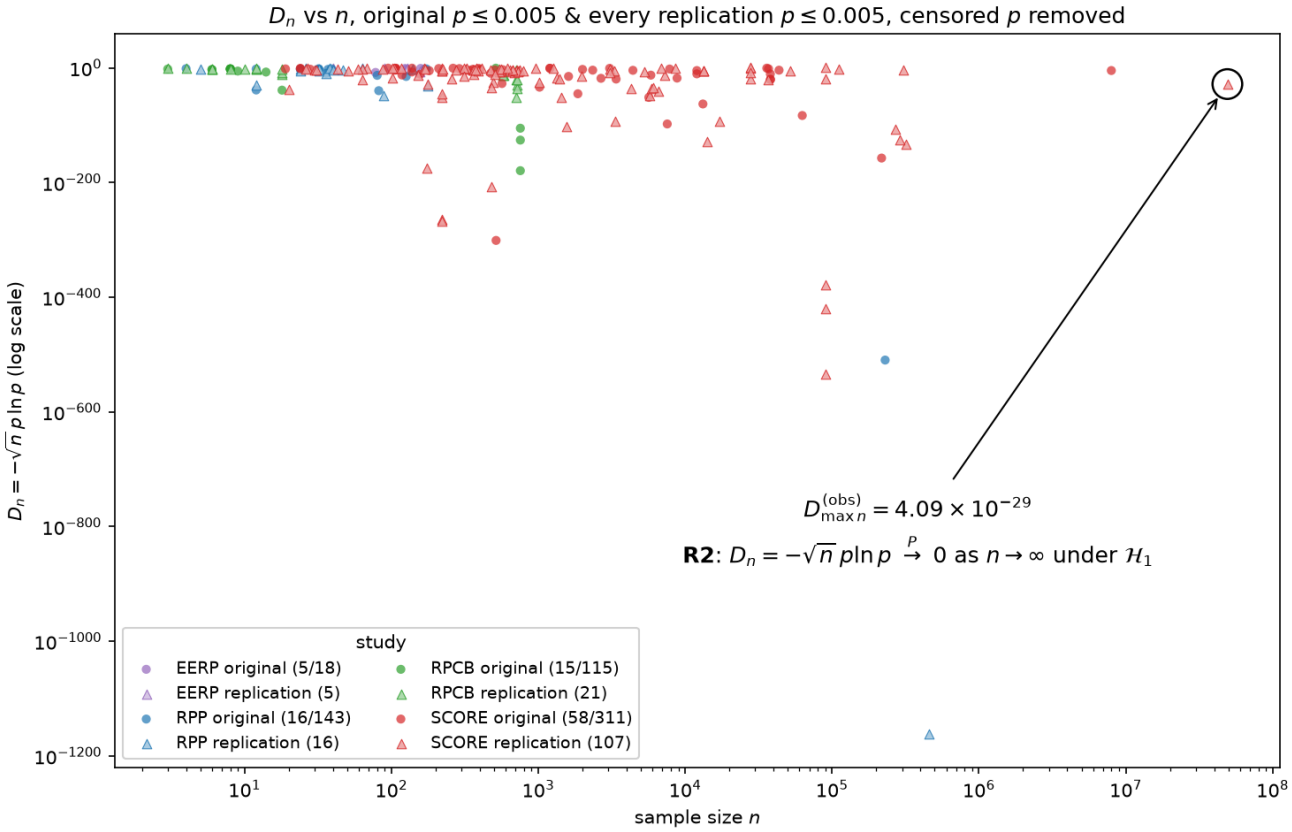


Figure S9: Realized values of $D_n = -\sqrt{n} \cdot p \ln p$, on a logarithmic scale, plotted against sample size n for the $\mathcal{H}_1$ subsets of the four replication projects (original $p \leq 0.005$ and every replication attempt $p \leq 0.005$). Circles denote original results and triangles replication attempts, with symbols colored by project. Legend entries give the number of plotted values; for original results this is shown relative to column 2 of Table S6. p-values reported as bounds are excluded. The circled value, $D_{\max n}^{(\text{obs})} = 4.09 \times 10^{-29}$, is the observed D_n at the largest sample size.

Appendix F: Previous posterior density comparison

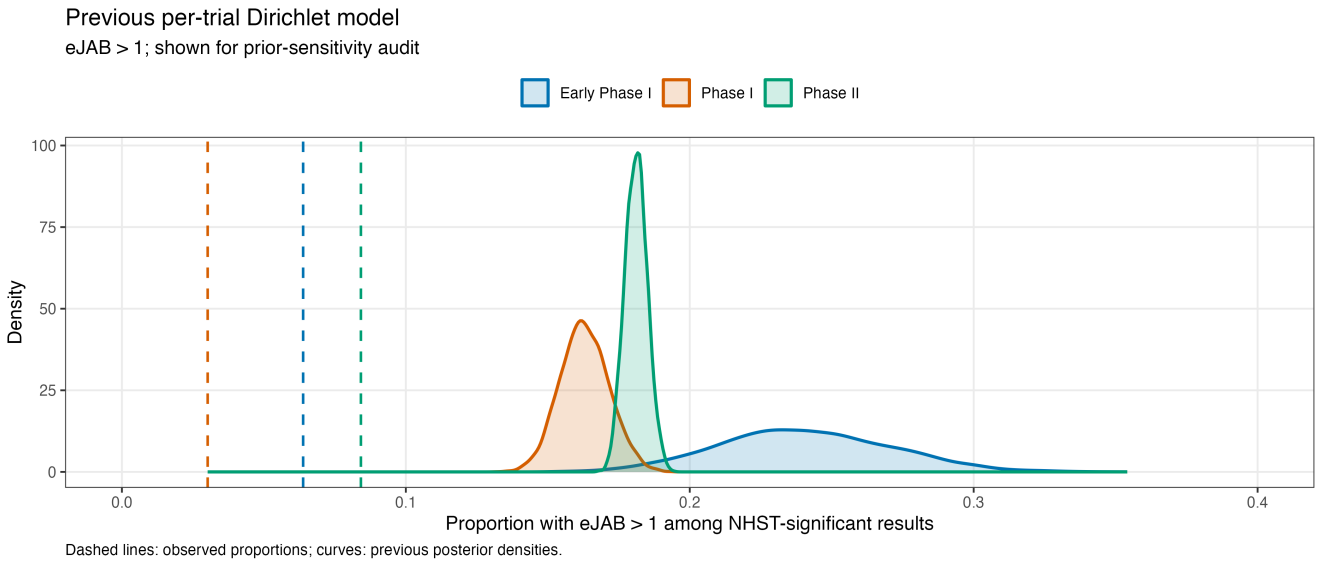


Figure S10: Previous per-trial Dirichlet$(1, 1, 1, 1)$ posterior densities for the proportion of NHST-significant records with eJAB > 1. Dashed lines mark the corresponding observed proportions. These densities are retained to document the prior-induced discrepancy and are not the estimates used in manuscript Figure 3B. The phase-specific priors and result-count weighting are described in Section C.8.

Appendix G: NHST significance-level profiles

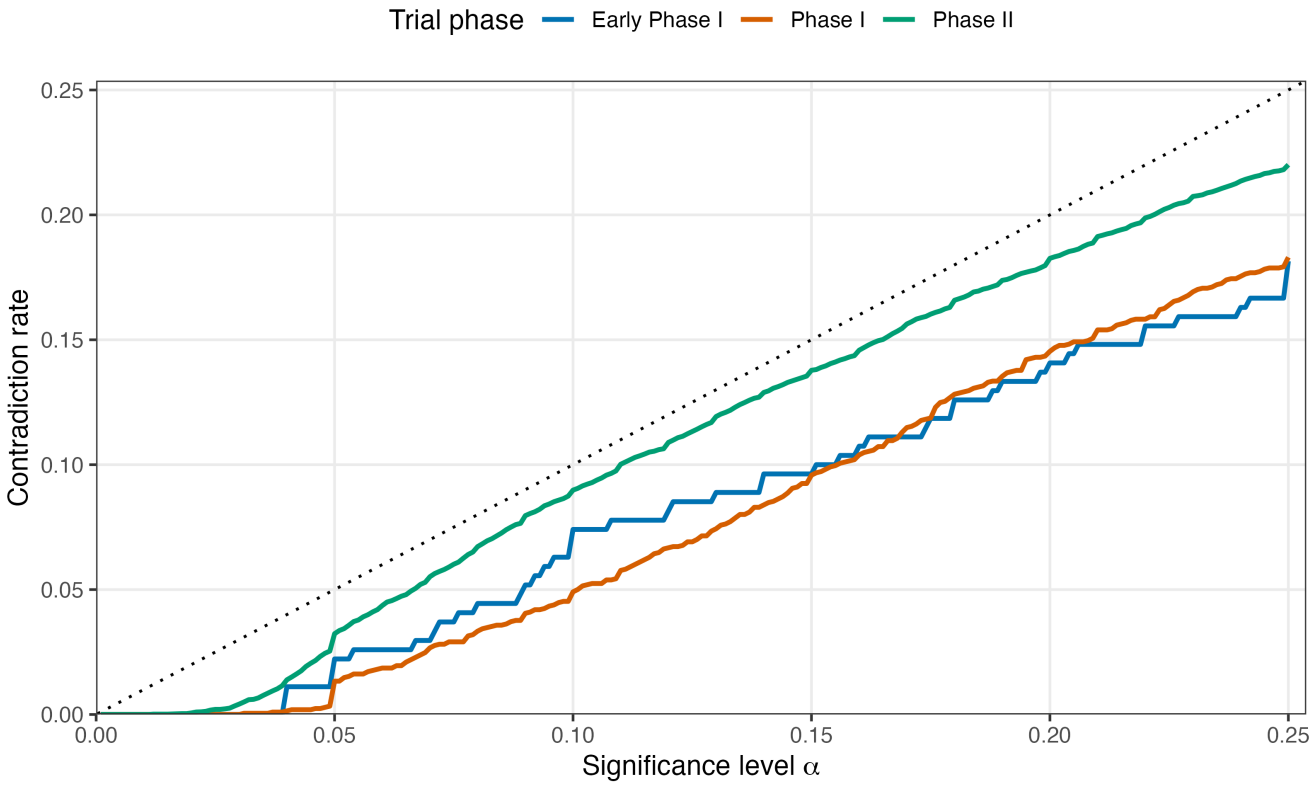


Figure S11: Overall Bayes/NHST contradiction proportions across nominal NHST significance levels, with separate curves for Early Phase I, Phase I, and Phase II. For phase g, the numerator $C_g(\alpha)$ counts eligible analyses with $p_{\text{NHST}} \leq \alpha$ and applicable eJAB > 1, and the denominator N_g is all eligible analyses in that phase. The dotted line marks equality with α. Reported values and upper bounds at .05 produce a jump; the curves describe screening frequency, not type I error rates. At $\alpha = .05$, these are the same overall proportions shown at $K = 1$ in manuscript Figure 3A. Figure 3B instead conditions on NHST significance and uses $S_g(.05)$ as its denominator.